

Where in England would Mobile Stroke Units be Cost-Effective and what are the Implications for Equity of Access: A model based Distributional Cost-Effectiveness Analysis

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Abstract

Background

Mobile Stroke Units (MSUs) have been shown to be an effective means to improve outcomes following stroke by increasing the rate and speed of thrombolysis. The consequences of MSU implementation in the UK's National Health Service have not been examined but is important because contextual geographical and organisational dependencies may influence effectiveness, cost-effectiveness, and equality of access. We aimed to develop a model to explore these issues across the ten English NHS Ambulance Services.

Methods

A previously developed model estimated the improvement to stroke outcomes through deployment of MSUs around the existing stroke service infrastructure. We estimated the net cost-effectiveness of MSUs by combining this with a model that estimates the lifetime health benefits and cost saving associated with these improvements in stroke outcomes. Inequality of health benefits was estimated in terms of absolute gains of the most and least deprived quintiles of populations. Possible MSU locations (either at a comprehensive stroke centre or an acute stroke unit) were compared in terms of cost-effectiveness and equity, both independently and combined in Distributional Cost Effectiveness Analysis.

Results

No locations were financially sustainable without the addition of the financial benefits derived from willingness-to-pay for health benefits. Over a patient's lifetime and from a secondary care perspective, more than one third of locations were cost-effective at a £20,000 willingness-to-pay threshold per Quality Adjusted Life Year, with each regional Ambulance Service having at least one cost-effective location for an MSU. However, choice of location of MSUs will also be influenced by a willingness to pay for health gains in the most deprived quintile compared to the general population, with all Ambulance Services having additional locations that could reduce inequity, at the cost of overall benefits.

Conclusions

MSUs have a heterogeneous impact on cost-effectiveness and equity of access of stroke care in the English NHS, with benefits mainly confined to localities with specific geographical and organisational characteristics. Commissioners can consider them as an option in the context of local service provision and resource availability along with other advances in prehospital care, but it cannot be assumed that they would be beneficial in all locations.

Background, Aims and Objectives

Stroke remains one of the leading causes of disability and death in the UK, with timely access to care being critical in improving outcomes and minimising long-term impacts. While stroke treatments have advanced significantly, reaching patients within the narrow therapeutic window remains challenging, especially in regions

with varied geography and resource availability. Mobile Stroke Units (MSUs)—specialized ambulances equipped with advanced stroke personnel with diagnostic and treatment capabilities—offer a potential solution by enabling immediate, pre-hospital stroke care. Based upon published trials in specific settings, European Stroke Organisation guidelines state that MSUs improve prehospital management of suspected strokes (1). A recent systematic review found the numbers needed to treat (NNT) to achieve an ‘independent’ outcome, defined as disability on the modified Rankin scale (mRS) of 0–2 at 90 days, ranged from 8 to 20 with some studies reporting no benefit depending on study type (2). The same review reported a NNT range of 9 to 30 for an ‘excellent’ outcome of mRS 0–1 at 90 days, with one study reporting no benefit. In both cases, the benefits were found in randomised control trials and not in observational studies suggesting that the value of an MSU is context dependant. Therefore, deploying MSUs within England’s healthcare system requires careful evaluation of their health benefits, cost-effectiveness, and equality/equity implications in the context of real-world service provision. Traditional Cost-Effectiveness analyses seek to maximise benefit regardless to where it accrues, assuming that quality adjusted life-years (QALYs) will be of equal value in all settings. However, by combining information about efficiency (cost-effectiveness) with information about inequities we draw on methods associated with Distributional Cost-Effectiveness Analyses (DCEA) that seek to acknowledge the trade-offs open to decision makers in different settings (3).

Our research builds on previous work that identified English ambulance services (AS) as the pragmatic way to implement MSUs if cost-effectiveness is acceptable, due to their comprehensive geographical coverage and established regional structures (4). The resulting pathway included MSU deployments by stroke teams joining emergency calls to relevant AS. The MSUs would then be crewed by a Radiographer, a Stroke nurse, a Paramedic, and a Care assistant with remote video review by a stroke specialist. Based on this model we developed a web-based application, co-produced with key stakeholders, which enabled preliminary analysis of stroke outcomes by various geographies and user chosen variables relevant to treatment delivery (5). Modelling was based on Lower Super Output Areas (LSOAs, English localities of approximately 1,500 population) and the primary outcome was the proportion of patients who were able to live independently at 6 months (mRS 0–2) predicted based upon times to Intravenous Thrombolysis (IVT) and Mechanical Thrombectomy (MT) (6, 7). Although the web application provides valuable insights, it does not encompass financial costs, long-term health outcomes, nor equity considerations. Recognizing the need for a more comprehensive tool, we developed a new model to assess these additional factors, informed by stakeholder engagement and grounded in real-world operational insights. This is important because initial analysis of MSUs by Laws et al focussed-on England as a whole (5). In this analysis benefits from MSUs were due to decreased times to treatment for thrombolysis and direct conveyance of eligible patients to thrombectomy-capable centres (in England termed ‘Comprehensive Stroke Centres’ (CSCs)). However, due to geographical factors, benefits varied at least five-fold between geographies (lower super output areas).

In England, the organisation of stroke care differs from the urban settings where MSUs have principally been tested, such as parts of mainland Europe and the urban United States of America (8–11). The AS cover a mix of urban, suburban, and rural areas and serve multiple stroke services with admission to the nearest one. Most are thrombolysis-only units, requiring secondary transfer to a regional CSC if thrombectomy is needed, but the arrangements differ considerably according to the underlying historical infrastructure and geography. The resulting inter-regional and intra-regional inequities in acute care access necessitate a tailored approach to MSU modelling which can account for specific geographical and organisational challenges faced across English ASs and recognise that there may be areas with higher and lower cost-effectiveness (12).

Our aim was to estimate the cost-effectiveness, and equity implications of MSUs in England. The objectives were:

- To extend the outputs of the models used in the web application to include details about costs, mortality, and morbidity estimates of MSU replacing standard ambulance care.
- To use the model's outputs to estimate if and where MSUs might be cost-effective at a local level within different ASs, and how they might impact health equity in terms of access to emergency ischaemic stroke treatment.
- To develop estimates on a per AS basis to understand how different situations might change both cost-effectiveness and equality/equity of access to timely stroke treatment and what the implications would be for equity from national versus regionally informed decisions.
- To describe how the implications of these findings might inform decisions about the use of MSUs in the English NHS.

Methods

Model structure and parameters

Building on an existing model of benefit by geography of MSUs (13) we added an estimated annual incidence of stroke based on the age, sex and deprivation profile of each Local Super Output Area (LSOA) and grouped these by ten English ASs (East of England AS NHS Trust, EEAST; East Midlands AS NHS Trust, EMAS; London AS NHS Trust, LAS; North East AS NHS Foundation Trust, NEAS; North West AS Trust, NWA; South Central AS NHS Foundation Trust, SCAS; South East Coasts AS NHS Foundation Trust, SECAmb; South Western AS NHS Foundation Trust, SWAS; West Midlands AS University NHS Foundation Trust, WMAS; and Yorkshire AS NHS Trust, YAS but not including the ambulance service on the Isle of Wight) (14) to derive an expected incidence per service. Our population was all incident stroke in England for one year (2021) when age and sex profiles were available by LSOA (15). The model uses age, sex and IMD to predict incidence (16). The benefits of MSU care over usual care were estimated using incidence information with the geographic benefits, in terms of decreased dependence and deaths projected over a lifetime using an existing model of life expectancy and secondary care costs (17). While an MSU may not be able to tailor their individual dispatches to only those patients who benefit the most, it is reasonable to assume, especially when capacity is considered, they can be limited to a geographic area that will maximise benefits. We labelled these limits on dispatches as configurations. Configurations were modelled as MSU being based in a location (either a CSC or acute stroke unit) and only being dispatched to strokes that occurred with a specified travel time of that location (30-, 45- or 60-minutes travel time). This means that, although the MSU is commissioned at a AS level, not all areas within the AS will necessarily be covered by an MSU. In addition, as stroke unit placement is dependent on population density, an MSU based at one location may be dispatched to patients who, under usual care, would be transported to a different stroke unit. Similarly, a patient who would, under usual care, be brought to a location may be outside the travel time for an MSU based at that location and therefore would still receive usual care. Therefore, the patient population for each MSU configuration is not identical to the current patient population for the stroke unit that the MSU is based at. Although the preference for the stakeholders who took part in the pathway development was for MSUs to have a dynamic location, this was beyond the scope of the model (4).

Treatment rates for any MSU were assumed to the maximum of either five per week or 20% of the MSU local incidence determined by the population covered by the specified maximum travel time. The model ignores the

number of dispatches and if they might occur outside the assumed working hours. For the model, local incidence was defined to be either 30, 45 or 60-minutes travel time to illustrate the trade-off between eligible population and the relationship between time to treatment and benefit. Benefit was defined as the mean benefit across all stroke patients who would benefit from treatment across the local incidence (i.e. there is no patient selection with the area covered by an MSU). To examine the heterogeneity of benefits and their distributions we modelled situating a MSU in each of the 26 English CSCs and Hyper Acute Stroke Units within each AS region. No other MSUs were assumed to exist within the AS and AS vehicles were assumed not to travel to patients located within other regions, however they would transfer patients to the nearest CSC or Stroke Unit if it were the nearest. Existing care would always transport patients to the nearest appropriate hospital. The perspective of the cost-effectiveness model was Secondary care and Ambulance Services. The time horizon was the patient's lifetime, and the comparator was usual care whereby patients are taken to their nearest unit and if that was not a CSC, the patient would be transferred by secondary ambulance if appropriate. The model is determinist; no attempt was made to capture or quantify uncertainties around incidents, treatment effectiveness, costs, or health outcomes. The model's purpose was to estimate how the siting of MSUs could affect their cost-effectiveness and how population health inequities might be affected.

Key Parameters to the model are shown in Table 1. In order not to bias the results in favour of MSU, a conservative approach was taken, in which variables for the standard pathway (such as ambulance response time) were assumed to take optimistic values. Other parameters, including those used in preexisting components of the model are shown in Supplementary Materials (supplementary table 1). All modelling was done using Stata (18) and Microsoft Excel (19).

Table 1
Model Parameters

Parameter	Pathway	Value (minutes)
Ambulance response time	Standard pathway	20
Time ambulance is on scene	Standard pathway	30
Hospital arrival to IVT time	Standard pathway	40
Hospital arrival to MT time (for in-hospital IVT + MT)	Standard pathway	90
Door-in to door-out (for transfer to MT)	Standard pathway	90
Hospital arrival to MT time (for transfers)	Standard pathway	60
MSU dispatch time	MSU pathway	15
MSU IVT time	MSU pathway	30
MSU on scene post IVT time	MSU pathway	0
MSU on scene (no thrombolysis given)	MSU pathway	15
Hospital arrival to MT time (for MSU arrivals)	MSU pathway	60
Scale factor for MSU travel speed	MSU pathway	1.00

Having previously reviewed the incorporation of inequalities in access to hospital reperfusion services, we use the term inequity because inequality in access acute stroke care is inherently unfair within the existing service structure (12, 20). Inequity was defined in terms of the difference in quality adjusted life years (QALY) gains between the most deprived and least deprived quintile of the population in each AS, described by the Index of Multiple Deprivation 2019 (IMD) (21). A positive difference indicates that most deprived quintile receive more benefits than the least deprived quintile and is therefore more equitable (or 'progressive'), and negative difference indicates that the least deprived quintile receives more benefits than the most deprived quintile and is less equitable (or 'regressive'). A separate analysis considered how inequity might be impacted at AS level if national IMD quintiles were used. The benefits of each configuration were calculated at the standard willingness-to-pay threshold for each QALY gain (£20,000) plus any cost savings from reduced secondary care usage minus the costs of the MSU. Dominated configurations (i.e., those configurations that have less benefits and contribute less to reducing inequity than other configurations) were removed. It was also assumed that decision makers would not choose configurations that provide less benefits to both most and least deprived quintiles but decrease inequity by reducing the most deprived by less than the least deprived. These configurations were also removed. The remaining configurations are presented with their benefits and equity implications and represent a range of possible configurations where a trade-off exists between maximising benefits and reducing inequity. The efficiency-equity trade-off is presented graphically using a distributional cost effectiveness plane.

For our analysis, IMD quintiles were estimated on a per AS basis, assuming that their decisions about MSUs were concerned with inequalities within their footprint rather than the national picture. We explore the consequences of this assumption.

Financial Costs

The costing perspective was Secondary care and Ambulance Services, and the costing year was 2023. The costs of setting up a MSU was assumed, based on expert opinion, to be £800,000 with a lifespan of 5 years. The non staff running costs were taken from Lund et al (2022) and were converted into pounds sterling and inflated to 2023 prices (22). Staff hourly costs were taken from the Unit Costs of Health and Social Care (23). It was assumed that a MSU would be run (and therefore staffed) for 37.5 hours per week, 52 weeks of the year (a total of 1950 hours per year). We are implicitly assuming that the MSU operates in addition to (rather than instead of) any other vehicles within the AS. It was assumed that, for both MSUs and the standard pathway, patients would be assessed by a stroke specialist. Therefore, there would be no cost difference for an MSU. Although the pathway development work found that telecare was a viable solution for MSUs, with no accurate costings, this was not included in the analysis (however, it is unlikely that this cost would be significant (24)).

Table 2
Cost components of an MSU

Cost Component	Cost (£)	Source
Annualised one year cost of setting up an MSU	177,185.00	Expert opinion
Non-staff operating costs of a MSU	65,367.00	Lund et al. (2022)(13)
Radiographer - Agenda for Change band 5	79,950.00	Unit Costs of Health and Social Care (14)
Stroke nurse - Agenda for Change band 7	144,300.00	Unit Costs of Health and Social Care (14)
Paramedic - Agenda for Change band 6	103,350.00	Unit Costs of Health and Social Care (14)
Care assistant - Agenda for Change band 3	25,674.00	Unit Costs of Health and Social Care (14)
Total Cost	595,826.00	

Analysis

We conducted a distributional cost-effectiveness analysis. This was based on an estimate of cost-effectiveness at £20,000 per QALY (current willingness to pay threshold used in the UK by NICE) and a secondary care and AS perspective and reported as an ICER. Utility values for independence and dependence following stroke were from the MRCLEAN study (25) and cost for future secondary care from the published Person Social Services Research Unit (26). Equity was measured in terms of the QALYs gained from the deployment of an MSU by the most deprived quintile over the least deprived. (measured by IMD). DCEA was presented on an Efficiency Equity plane.

Sensitivity analysis

Three sensitivity analyses were run to account for differences within the populations. Firstly, the analysis was repeated using national IMD quintiles, to account for any differences in deprivation between ambulance services. Secondly, the analysis was rerun only from the perspective of the ambulance services. That means the potential cost savings from reduced secondary care usage was not included. Although in the base model, treatment rates for any MSU were assumed to the minimum of either five per week or 20% of the MSU local incidence determined by the population covered by a maximum travel time, as a sensitivity analysis, we calculated the minimum number of patients needed to be seen for there to be a cost-effective configuration for each AS.

Results

Deprivation differed between ambulance service (supplementary table 2). Over one third of LSOAs in the NEAS and NWAS areas were classified as being in the most deprived quintile nationally, while under 10% of LSOAs in SCAS and SECamb were classified as being the most deprived. We predicted the number of strokes by AS, with NWAS having the largest number of strokes (9,677) and NEAS having the lowest (3,731). Although deprivation is associated with increased risk of stroke, the population aged 45 to 89 increases as IMD quintile increases, leading to a higher incidence of strokes in the least deprived quintiles (14).

Cost-effectiveness

Most possible configurations included were not cost-effective, with only 108 out of 300 having an ICER below £20,000. Each AS had at least three cost-effective configurations (LAS), with EEAST having the most options (19) (Table 3). All possible configurations (300 configurations, see supplementary tables) lead to cost savings from reduced secondary care usage, although in no configuration was this larger than the costs of the MSU. In all ASs combined, the most cost-effective configuration was St Peter's Hospital (in Surrey, part of SECamb) within 30 minutes travelling distance, with a total QALY gain of 37.95 and an Incremental Cost Effectiveness Ratio (ICER) of £13,403.41. If the cost saving from reduced secondary care usage are removed, there is little effect on the cost-effectiveness of MSUs, with the most cost-effective configuration for each ambulance service remaining the same.

Table 3
Cost-effectiveness of MSUs, with and without cost savings from reduction in secondary care usage, by ambulance service

Ambulance Service	Number of cost-effective configurations	Most cost-effective configuration (base location and maximum travel time)	Number needed to treat	Total lifetime QALY gains	Total Costs including secondary care cost savings	ICER with secondary care cost savings (£)	ICER without secondary care cost savings (£)
EEAST	19	Luton and Dunstable Hospital within 30 minutes	34	36.62	£512,264	14,000	16,300
EMAS	11	Leicester Royal Infirmary within 30 minutes	33	35.12	£510,638	14,500	17,000
LAS	3	Princess Royal University Hospital HASU within 30 minutes	40	32.77	£514,993	15,700	18,000
NEAS	10	Sunderland Royal Hospital within 30 minutes	42	30.57	£520,629	17,000	19,500
NWAS	12	Stepping Hill Hospital within 30 minutes	38	33.93	£512,087	15,100	17,600
SCAS	8	Royal Berkshire Hospital within 45 minutes	37	35.23	£518,844	14,700	17,000
SECamb	18	St Peter's Hospital within 30 minutes	34	37.95	£508,693	13,400	15,700
SWAS	9	Royal Devon and Exeter Hospital within 45 minutes	38	32.3	£532,241	16,500	18,400
WMAS	9	New Cross Hospital within 30 minutes	36	35.2	£511,898	14,500	16,900

Ambulance Service	Number of cost-effective configurations	Most cost-effective configuration (base location and maximum travel time)	Number needed to treat	Total lifetime QALY gains	Total Costs including secondary care cost savings	ICER with secondary care cost savings (£)	ICER without secondary care cost savings (£)
YAS	9	Calderdale Royal Hospital within 30 minutes	37	35.23	£510,964	14,500	16,900

Equity

All AS had at least one site for an MSU that improved the health of the most deprived quintile of their constituent population (supplementary Fig. 1). The estimates of the effect on inequity are shown in the Supplement and a subset are included in the following subsection about DCEA. The most extreme case was in SECamb. Our model showed that, depending on the choice of site, either the most deprived within SECamb's catchment could gain over ten QALYs more than the least deprived or the least deprived could gain over ten QALYs more than the most deprived.

Distributional Cost Effectiveness

Net Benefit with effect on equity was plotted on a four-quadrant graph. Equity is measured on the x-axis where zero represents the circumstance where the site of an MSU makes no difference to the distance between the health outcomes of the least and most deprivates quintiles, a positive value represents a reduction in inequality and a negative value an increase. The y-axis represents Net Benefit (in this case at £20,000 per QALY) (27); a positive number means additional benefits beyond the £20,000 per QALY threshold, and a negative value means the site would not be considered cost-effective. Net Benefit is also referred to in terms of efficiency because a high Net Benefit implies higher efficiency.

These planes are shown for each service in Fig. 1 the red circles represent choices made based on deprivation quintiles of the AS's constituency. The Methods section describes assumptions we have made about decision makers preferences which have resulted in the removal of most configurations. Which configuration is the optimal choice is dependent on the decision maker's willingness-to-pay for equity gains or forgo them in some circumstances. For example, the most cost-effective configuration (i.e., the configuration with the most benefits) for SECamb is St Peter's Hospital within 30 minutes travelling distance. However, this configuration provides 9.05 more QALYs to the least deprived quintile compared to the most deprived i.e. it is regressive in terms of health benefits. Comparatively, Maidstone District General Hospital within 45 minutes travelling distance provides £60,746 less in total benefits but is progressive, delivering 4.99 more QALYs to the most deprived compared to the least deprived.

In panel G, there are three options for SWAS in order of decreasing effectiveness and increasing equity, shown by the red circles (ranging from the Royal Devon and Exeter Hospital within 45 minutes travelling distance to the Royal Cornwall Hospital within 60 minutes travelling distance). The most efficient option is the Royal Devon and Exeter Hospital within 45 minutes travelling distance, with a benefit of £113,759. However, by choosing this site,

the most deprived quintile gains 0.14 fewer QALYs than the least deprived, increasing inequity. If the MSU were instead situated in Musgrove Park Hospital within 45 minutes travelling distance, the overall benefit is reduced by £76,534 (roughly the cost equivalent of 3.8 QALYs ignoring the small component of Net Benefit resulting from reduced future healthcare expenditure) but equity is increased, with the most deprived gaining 2.47 more QALYs than the least deprived. The option which reducing inequities the most is Royal Cornwall Hospital within 60 minutes travelling distance, where the most deprived quintile gains 6.81 more QALYs than the least deprived. However, this gain in equity comes with a reduction in benefits of £198,718 when compared to the Royal Devon and Exeter Hospital within 45 minutes, the cost equivalent of 9.94 QALYs. This reduction in benefits is large enough that the Royal Cornwall Hospital within 60 minutes is not considered cost-effective at £20,000 per QALY.

However, a decision to site an MSU at the Royal Cornwall Hospital within 60 minutes rather than at the Royal Devon and Exeter Hospital within 45 minutes would imply that gaining one more QALY in the most deprived quintile compared to the least deprived indicates a willingness to pay of £28,600 per QALY in the most deprived quintile, based on the ratio between QALYs lost from net benefit reduction and equity gained. This is calculated by dividing the net benefit reduction by the difference in equity (£198,718 divided by 6.95 QALYs), ignoring any reduction in cost saving, which are a relatively small part of benefits.

The net benefit to each configuration was, in the main, driven by overall QALY gains. Low net benefit for a configuration was due to one of two reasons: (1) the local population and/or incidence rate was too low for the benefits of the MSU to outweigh the cost or; (2) the population served by the MSU configuration already has good access to treatment and the benefit per stroke of the MSU is too low to outweigh the cost. It is also worth noting that no AS had a CSC as the candidates for location based on maximising benefits alone. Two ASs (LAS and NEAS) had CSCs that delivered the biggest reduction in inequity. Therefore, some CSC locations in some ASs may be candidates based on decision makers' willingness to trade net benefit for equity gains.

National versus sub-national commissioning sensitivity analysis

The effect on equity can differ significantly if national IMD quintiles are chosen (the blue triangles in Fig. 1), rather than quintiles based on each AS's footprint (shown in Fig. 1 as red circles). Although for the SWAS area alone, Royal Devon and Exeter Hospital within 45 minutes travelling distance has little impact on equity (with a difference between the most and least deprived quintiles of -0.14), the SWAS area has less of the most deprived populations than the national average, with only 11% of LSOAs within the SWAS area being in the most deprived quintile nationally. If decision makers were concerned more with national inequities and used quintiles based on national quintiles, Royal Devon and Exeter Hospital within 45 minutes travelling distance increases inequity significantly, with the most deprived national quintile gaining 4.6 less QALYs than the least deprived national quintile. The situation in other ambulance service, such as NWAS, is different, where MSU configurations that have little effect on equity within the region can have a large impact of inequity nationally. Within some ASs, there exist configurations which increase equity (at the cost of overall benefits) within the AS's footprint (i.e., the most deprived of the population that live within the AS gain more than the least deprived) but are dominated when deprivation is considered on a national level. For example, University Hospital Aintree within 30 minutes is cost-effective and increases equity using AS level measures of deprivation (albeit at a reduced benefit), when using national level measures of deprivation, it is dominated by Fairfield General Hospital within 30 minutes, which provides more benefit and more equity.

Minimum number of patients required

We reduced the maximum number of confirmed ischaemic stroke patients a MSU could see in a year to find the minimum number of patients that still resulted in a cost-effective service (Table 4). For five ASSs, the most effective configuration does not change as the maximum number of patients is reduced (LAS, NWAS, SCAS, WMAS and YAS). However, for EEAST, EMAS, NEAS, SECamb and SWAS, the most cost-effective option at 250 patients/year changes as the number of patients falls. There is a trade-off between maximising benefits per patient and maximising benefits for the population as a whole. Many configurations have a relatively small footprint and, although per patient they may be very effective, without the population necessary to fully utilise the MSU, they are either not the most cost-effective option or not cost-effective at all. For example, for EEAST, Watford General Hospital within 30 minutes provides the most benefit per patient. However, it lacks the population to see 250 stroke patients per year, therefore Luton and Dunstable Hospital within 30 minutes is the most cost-effective option. As the number of patients treated falls, Luton and Dunstable Hospital within 30 minutes cost-effectiveness decreases faster than Watford General Hospital within 30 minutes until, at 173 patients, Watford General Hospital within 30 minutes is the only cost-effective configuration. For configurations that increase inequity, lowering the number of patients seen will reduce this increase. Conversely, for configurations that decrease inequity, lowering the number of patients seen also lowers this decrease.

Table 4
Minimum number of stroke patients required to be cost-effective, by ambulance service

Ambulance Service	Minimum patient numbers for one configuration to be cost-effective	Configuration	Total QALY gains	ICER (including secondary care cost savings)	Change in equity from 250 patients to minimum patients required
EEAST*	173	Watford General Hospital within 30 minutes	27.06	19,700	0.15
EMAS*	170	Northampton General Hospital within 45 minutes	26.79	20,000	0.92
LAS	208	Princess Royal University Hospital HASU within 30 minutes	27.26	19,400	1.41
NEAS*	204	University Hospitals of North Tees and Hartlepool within 30 minutes	26.65	19,900	0.01
NWAS	196	Stepping Hill Hospital within 30 minutes	26.60	19,900	0.74
SCAS	191	Royal Berkshire Hospital within 45 minutes	26.92	20,000	1.98
SECAmb*	173	Maidstone District General Hospital within 30 minutes	26.88	19,900	-1.43
SWAS	185	Great Western Hospital Swindon within 45 minutes	26.92	20,000	0.25
WMAS*	190	New Cross Hospital within 30 minutes	26.76	19,900	0.39
YAS	189	Calderdale Royal Hospital within 30 minutes	26.64	20,000	0.30
* the most cost-effective option at 250 patients per year does not have the minimum patients required					

Discussion

This is the first economic study we are aware of into the deployment of MSUs at a national level and the first to undertake a DCEA. We extended a previously developed geographic model to include incidence and have shown that although MSUs are not cost-effective when widely deployed, they could be (in certain configurations with stated assumptions) an effective way to reduce long term healthcare costs and improve patient outcomes through the more rapid delivery of reperfusion treatments for acute ischaemic stroke, although this would also require addressing other implementation challenges (4). Our estimated NNTs are higher than previously

published studies, and we have demonstrated how the siting of an MSU could affect its efficiency, in terms of Net Benefit, and the health-related inequity of access to stroke care, defined by IMD. Cost-effectiveness was driven by the value of estimated future healthcare QALYs over a lifetime offsetting the costs of the MSU. Subsequent reduced healthcare costs beyond the initial episode of acute stroke care played a minor role in determining cost-effectiveness. Our model showed that within any given regional AS there were options to site MSUs in ways that could increase or decrease inequalities in access to care and resulting health outcomes, which also vary according to the size of the population that would be best served. We also demonstrated how commissioning decisions at a local AS level could either support or conflict with national priorities in terms of policies to reduce health inequalities. This is an issue worthy of further consideration when identified uncertainties about responsibility for commissioning and funding for MSUs are decided (5).

We estimated that siting MSUs at CSCs was, in most cases, likely to be less cost-effective than other locations within each AS, as these patients already have near and direct access to thrombectomy. This was largely driven by smaller health benefits but is inconsistent with deployment models in other locales such as Germany (28). However evidence from elsewhere concluded “there are few scenarios where MSU transport predicts substantially superior outcomes to the mothership method when the MSU is dispatched from the endovascular therapy center” (29). This is consistent with our previous findings that benefits arise predominantly when an MSU means that extended travel and delays associated with a Drip- and-Ship model are avoided because patients do not require secondary transport to CSCs. The ideal setting for an MSU appears to be a relatively large and deprived population distant from a CSC, which would otherwise require a secondary transfer for thrombectomy.

Our model has not considered the number of dispatches a MSU undertakes, only the number of confirmed ischaemic stroke patients eligible for treatment. While we have limited the number of patients eligible for treatment to a maximum of 250 per year, a MSU will almost certainly be dispatched significantly more often, often to patients who are either ineligible for IVT or MT, to patients with intracerebral haemorrhage or to stroke mimics. However, modelling this would require assumptions about the sensitivity and specificity of the MSU to identify which patients are likely to be eligible for treatment before dispatch. When considering the implementation of a MSU, attention must be given to the volume of dispatches the MSU can be expected to perform, and how this is perceived relative to the demands on other AS resources. If the number of eligible patients is too low, either due to low levels of dispatches or poor identification of eligible patients, the MSU configuration may not be cost-effective. In addition, the working pattern of a MSU may make a significant difference. We have made no assumptions within our model but, with 60% of strokes occurring between 6am and noon (30), an MSU operating within these hours seven days/week may have a large enough population of potential patients to be cost-effective while an MSU operating outside these hours or on fewer days may not.

Our assumptions are based on the best data available and an existing body of previous models. However, there are several limitations to the study based on the assumptions made about how and when MSUs were deployed. It is unlikely, for example, that deployment decisions would be made based on travel times. More likely decisions would be based on other factors as well including the patients’ circumstances. These decisions would then impact the effectiveness and subsequent cost-effectiveness of the MSU. It is unlikely that the effects on equity would be directly affected with decisions to deploy possibly being contingent on geography, but not on socio-economic circumstances. Our assumption about perspective of the financial consequences of stroke likely result in underestimates of savings as primary care, social care and societal costs have not been included, with improved outcomes likely to reduce demand. However, uncertainty around any estimates increases as more

categories of costs are included and our decision to exclude uncertainty from our analyses was made to highlight one source of variation: the location of the MSU.

Another limitation of our research was the choice of comparator as usual care. The recent evolution in stroke treatments, particularly the introduction of mechanical thrombectomy, has expanded the scope of acute stroke care. Thrombectomy is highly time-sensitive and available in England only at specialised CSCs, requiring that some patients bypass, or are transferred from, closer hospitals in favour of CSCs equipped to perform the procedure. Emerging technologies that enable accurate pre-hospital diagnosis could play a role in directing patients to appropriate facilities (29, 31). MSUs providing pre-hospital stroke assessment, could facilitate this process; however, they represent only one part of a changing system of stroke care pathways which could potentially reduce the benefits of MSUs. It may be that other modifications to the hyperacute stroke pathway may deliver similar population benefit – for example, significant reductions in door-in-door-out times for referring primary stroke centres may diminish the advantage conferred by MSU bypassing for patients eligible for MT, which is one of the principal contributors to the overall benefit from MSUs. There may be alternative, and cheaper, methods of achieving similar benefits e.g. through adopting a policy of holding the conveying ambulance at the primary stroke centre while evaluation for onward transfer for MT is completed. Furthermore, implicit in our model is the assumption that the model pathway delivers the outcomes predicted by the model. The scope of the model was limited to one MSU per AS. Although for some ASs (particularly geographically dispersed areas such as SWAS), there is the possibility of implementing two or more MSUs, the complexity of any overlapping units within and beyond AS footprints was beyond the scope of this study. Similarly, we have not modelled any potential clinical benefits, which may be considerable, from the earlier treatment of patients with acute intracerebral haemorrhage by MSUs (32), nor from the earlier management of stroke mimics. In some instances, in which the cost-effectiveness is marginal, any additional benefits derived from these groups of patients could be decisive.

The modelled benefits of MSUs were derived from reduced times to treatment. Some studies have reported that over 97% of eligible patients were treated by MSUs compared to only 80% in usual care (33). Including local treatment rates to estimate potential additional benefits was beyond the scope of this model, but significant variation in thrombolysis rates between centres in England does exist (34) and could mitigate the benefits of ambulance redirection based on MSU assessment.

For policy makers our research shows that MSUs can play a role in increasing access to timely treatments, but this role is determined by regional factors and MSU location. Importantly, unlike some other options, MSUs have the potential to directly impact health inequities for better or worse. However, further research is needed into their implementation, in the context of improvements in other pre-hospital triage approaches, service quality improvement and optimal configuration.

Conclusion

Our nationwide modelling study has shown that only in certain localities in the English NHS would MSUs represent a cost-effective option for reducing disability from acute ischaemic stroke at the customary willingness-to-pay threshold of £20,000/QALY, principally through the earlier delivery of reperfusion treatments, including redirection to thrombectomy-capable centres. Cost-effectiveness, and the impact of MSUs on existing health inequities, are sensitive to location and generally involve siting a MSU in populous areas away from the thrombectomy-capable centres, such that within each of the 10 English Ambulance Services there may exist

several locations where the introduction of an MSU in place of usual care achieves both objectives of being cost-effective and reducing health inequity.

Abbreviations

MSU - Mobile Stroke Units

NNT – Numbers needed to treat

mRS - modified Rankin scale

DCEA - Distributional Cost-Effectiveness Analyses

AS - ambulance services

NHS – National Health Service

IMD - Index of Multiple Deprivation

EEAST - East of England AS NHS Trust

EMAS - East Midlands AS NHS Trust

LAS - London AS NHS Trust

NEAS - North East AS NHS Foundation Trust

NWAS - North West AS Trust

SCAS - South Central AS NHS Foundation Trust

SECAmb - South East Coasts AS NHS Foundation Trust

SWAS - South Western AS NHS Foundation Trust

WMAS - West Midlands AS University NHS Foundation Trust

YAS - Yorkshire AS NHS Trust

LSOA – Lower Super Output Area

CSC - Comprehensive Stroke Centres

QALY - Quality adjusted life years

ICER - Incremental Cost Effectiveness Ratio

IVT - Intravenous Thrombolysis

MT - Mechanical Thrombectomy

Declarations

Clinical trial number not applicable

Ethics approval

Not applicable

Consent for publication

Not applicable

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests

S.M., J.S., L.S., M.A., A.L., L.M., G.M., L.P., D.P., D.W., P.M., report no competing interests. C.P. reports that they receive salary support from NIHR Programme Grant for Applied Research. G.F. reports that his organisation receives payments from other NIHR grants, consulting fees from CSL Behring stroke trial design and Astra Zeneca interview on stroke management, payment for lectures from Bayer Cardiovascular meeting with Primary Care Cardiovascular Society, and meeting expenses from Novartis. G.F. also participated on a Data Safety Monitoring Board or Advisory Board for CSL Behring Stroke Advisory Board (personal payment), Bayer Stroke trial advisory Board (payment to institution) and Advance Accelerator Applications round table event on radioligand services (personal payment). M.J. reports that they are an unpaid Trustee for the Stroke Association. P.W. reports receiving institutional unrestricted educational grants from Stryker, Penumbra and Metronic and that they participate on a Data Safety Monitoring/Advisory Board for MR CLEAN NO IV, TENSION, PROTECT U (all unpaid).

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P.M., J.S., M.A., L.S., M.J., G.F., G.M., P.W., C.P., D.P., and D.W. contributed to study conception and design. S.M., P.M., J.S., L.M., M.A., and A.L. contributed to data collection and analysis. All authors contributed to interpreting the data. S.M. and P.M. drafted the manuscript. All authors reviewed and approved the final manuscript.

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Figures

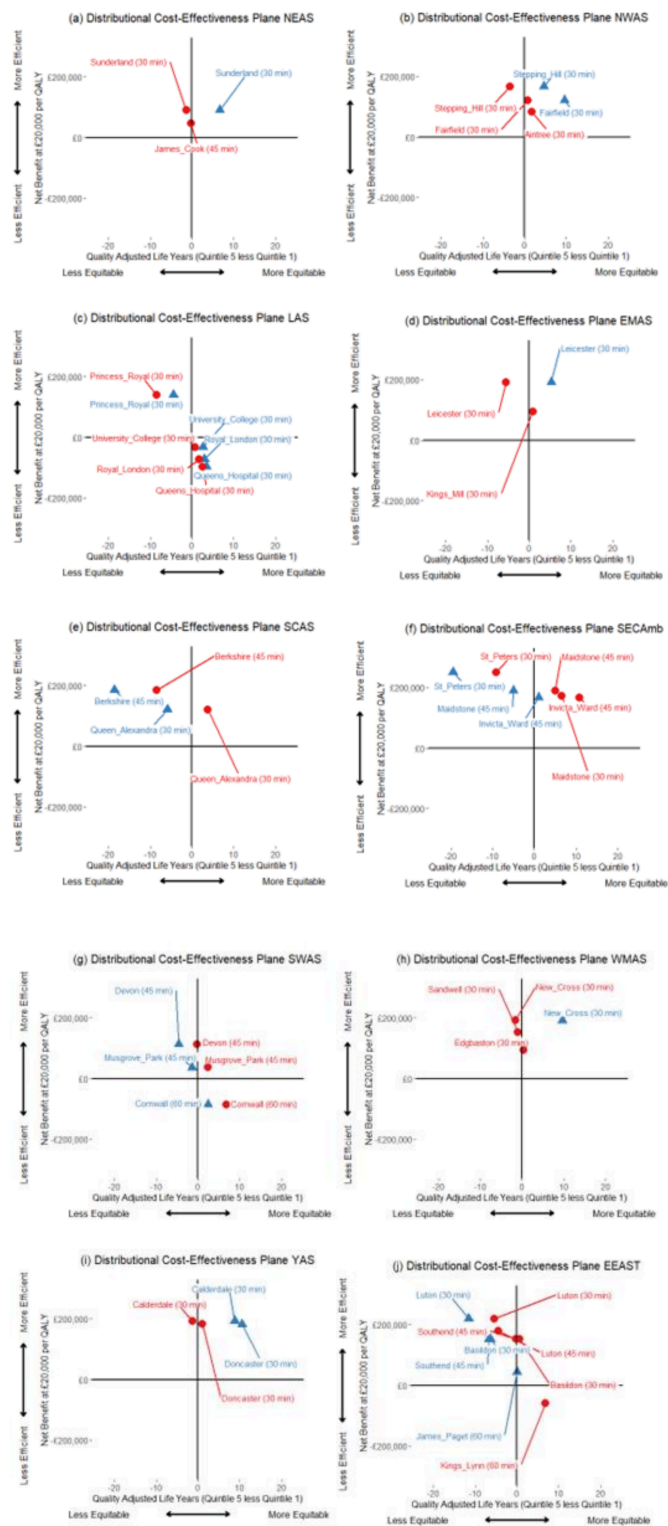


Figure 1

Distributional Cost Effectiveness planes by ambulance service

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